Lesson 8: Resonance

ChatGPT edition — mechanical evidence before electrical analogy

Frequency, exchange, and damping

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

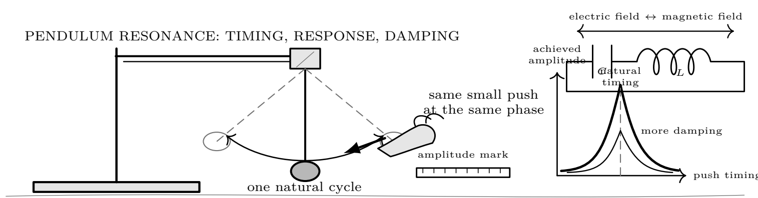
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- String pendulum and small mass
- Stand, ruler, masking-tape index
- Cardstock damping vane
- Stopwatch used away from apparatus
- Lab partner to call cycles

Predict first

Three equal pushes arrive early, at the same phase, or late. Rank the final amplitudes, sketch the response curve you expect, and state what result would change your mind.



Investigate

1. **Find the natural timing.** Pull the mass aside by 3 cm; release without pushing. Time ten complete cycles from one same-direction crossing to the eleventh. Repeat three times. Calculate

$$T = \frac{t_{10}}{10}, \qquad f_0 = \frac{1}{T}.$$

   Record all trials rather than choosing the neatest one. If the three periods differ by more than 5%, inspect the pivot, release, and cycle count before continuing.

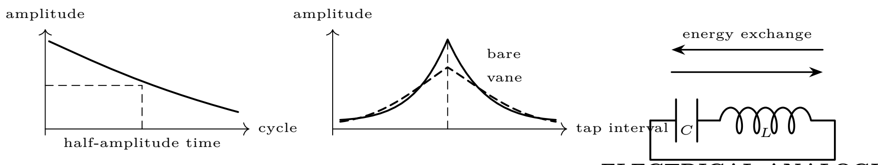
2. **Make push size repeatable.** Put a tape index beside the lowest point. A partner gives one light sideways tap only as the mass crosses that index in the chosen direction. Rehearse with the pendulum still: same finger, same travel, no following-through.
3. **Sweep the drive timing.** Reset to the same 3 cm release before each run. Give ten equal taps at intervals

$$0.70T, \quad 0.85T, \quad 1.00T, \quad 1.15T, \quad 1.30T.$$

   After tap 10, read maximum displacement against the ruler. Stop a run if the mass or string approaches the stand.

tap interval	$0.70T$	$0.85T$	$1.00T$	$1.15T$	$1.30T$
amplitude (cm)					
observed phase					

4. **Measure damping.** At the best interval, remove the taps and count cycles for the amplitude to fall from A_0 to about $A_0/2$. Attach the cardstock vane and repeat. Then repeat the five timing points with the vane, using the same release and ten taps.



FREE DECAY   RESPONSE SWEEP   ELECTRICAL ANALOGY

Explain from the evidence

Decide from the curve. A resonance claim needs more than "it got large." The response must rise toward one drive interval and fall away on both sides, with the peak near the independently measured T. The taps supply energy; phase determines whether each tap adds to or opposes the motion. Resonance selects efficient transfer. It creates no energy.

Pattern	Diagnose before interpreting
All points alike	Tap timing or tap strength was not controlled; widen the timing sweep.
Peak shifted	Recalculate T; check whether the string length or mass position changed.
Jagged left/right points	More trials are needed; the ruler reading or phase call is inconsistent.
Amplitude grows without bound	Stop: geometry changed, the stand is moving, or the safe-motion limit was missed.
Vane peak lower and wider	Expected damping signature: more energy lost each cycle and poorer frequency selectivity.

For an ideal electrical resonator,

$$f_0 = \frac{1}{2\pi\sqrt{LC}}.$$

Capacitor electric-field energy and inductor magnetic-field energy trade back and forth; resistance dissipates part of that energy on every cycle. This is an analogy supported by the shared timing, response-peak, and damping patterns—not proof that a pendulum contains an electrical circuit.

Change one thing

Final proof. On one graph, plot amplitude against t_{tap}/T for the bare pendulum and vane. Mark 1.00. Then answer:

1. Does each curve have an interior maximum rather than merely increasing from left to right?
2. Is the bare peak within one tested timing step of $1.00T$?
3. Does the vane shorten the half-amplitude decay time and reduce or broaden the peak?
4. Which observation rules out "all repeated pushes work equally well"?
5. What measurement uncertainty could move your conclusion?

Accept the resonance claim only if the independent period, two-sided response peak, and damping change agree. Otherwise report "not yet demonstrated" and name the next controlled repeat.

Evidence record

Prediction	______	One change	______
Observation	______	Claim + because	______

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.